

Figure 1: HBase table

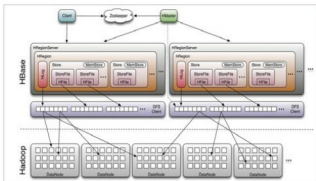


Figure 2: HBase architecture

row updates are atomic, no matter how many row columns constitute the row level transaction.

HBase is made up of an HBase master node orchestrating a cluster of one or more region servers as workers. The HBase master is responsible for bootstrapping the first time install, for assigning regions to the registered region servers and for recovering region server failures. HBase depends on Zookeeper and, by default, it manages a Zookeeper instance as the authority of the cluster state, although it can be configured to use an existing Zookeeper cluster instead.

Installation

To install HBase, download the stable release from an Apache Download Mirror (the link is <http://www.apache.org/dyn/closer.cgi/hbase/>) and unpack it on your local file system, by running the command `tar xvf hbase-0.96.1.1-hadoop2-bin.`

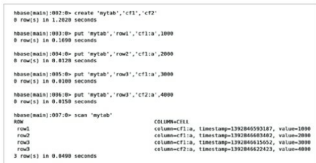


Figure 3: Inserting records in HBase

Data file:
1, India, Haryana, Chandigarh, 2009, April, P1, 1, 5
2, India, Haryana, Ambala, 2009, May, P1, 2, 10
3, India, Haryana, Panipat, 2010, June, P2, 3, 15
4, United States, California, Fresno, 2009, April, P2, 2, 5
5, United States, California, Long Beach, 2010, July, P2, 4, 10
6, United States, California, San Francisco, 2011, August, P1, 6, 20

Figure 4: Data file

tar.gz. HBase needs to know where Java is installed, so set `JAVA_HOME` in the environment variables. Start HBase by running the command `.bin/start-hbase.sh` from the HBase directory, and open the HBase shell by running the command `.bin/hbase shell` from the HBase directory. In the HBase shell, try the commands given below to create a table.

```
create 'mytab', 'cf1','cf2'
put 'mytab', 'row1', 'cf1:a', 1000
put 'mytab', 'row2', 'cf1:a', 2000
put 'mytab', 'row3', 'cf1:a', 3000
put 'mytab', 'row3', 'cf2:a', 4000
scan 'mytab'.
```

Congrats! You have your first code with HBase. Now let's look at what happened.

Create: Creates a new table identified by 'mytab' and Column Family identified by 'cf1'
Put: Inserts a new record into the table with row identified by 'row1.'

Scan: Returns the data stored in the table

Get: Returns the records matching the row identifier provided in the table

Now let us see how to create and populate the HBase table using Java. This data has to be inputted into a new HBase table to be created through JAVA APIs.

The following column families have to be created: "sample,region,time,product,sale,profit".

The column family region has three column qualifiers: country, state, city.

The column family time has two column qualifiers: year, month.

```
package Hbase;
import java.io.BufferedReader;
import java.io.File;
import java.io.FileReader;
import java.io.IOException;
import java.util.StringTokenizer;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.hbase.HBaseConfiguration;
import org.apache.hadoop.hbase.HColumnDescriptor;
import org.apache.hadoop.hbase.HTableDescriptor;
import org.apache.hadoop.hbase.client.HBaseAdmin;
import org.apache.hadoop.hbase.client.HTable;
import org.apache.hadoop.hbase.client.Put;
import org.apache.hadoop.hbase.util.Bytes;
```